



WISER × Womanium 2025 Global Quantum Program

Project 3

Quantum Algorithm as a PDE Solver for Computational Fluid Dynamics (CFD)

Eigenflow



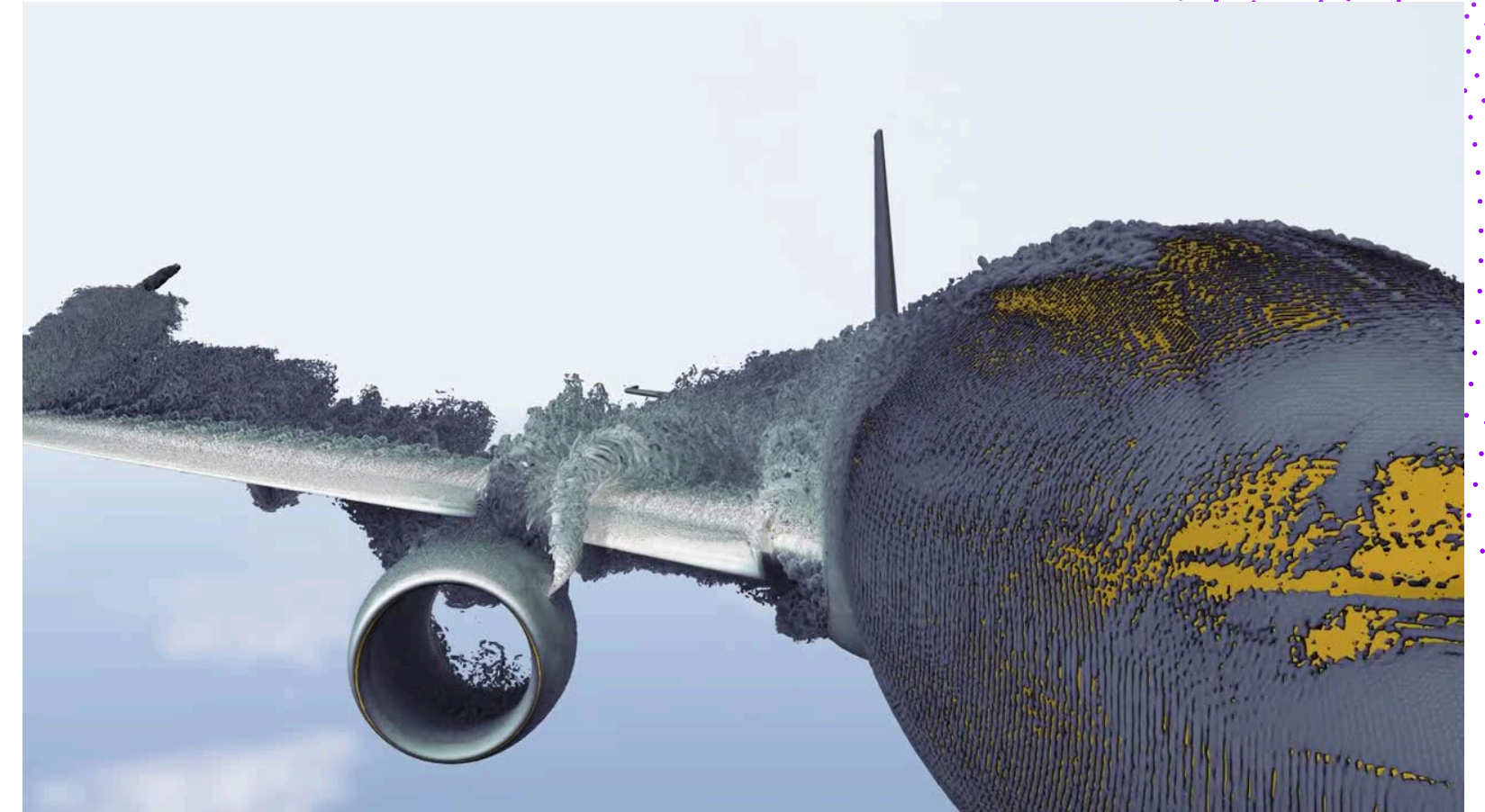
The Challenge: Computational Fluid Dynamics Limitations

Classical methods struggle with:

- Numerical diffusion at shock fronts
- Poor conservation properties
- Exponential scaling for complex flows

Quantum methods struggle with:

- Stable and noiseless qubits



Animation 1: Courtesy of Luis

$$\text{PDE: } \frac{\partial u}{\partial t} + \frac{u \partial u}{\partial x} = \frac{v \partial^2 u}{\partial x^2}$$

Domain: $x \in [0, 1]$

IC: Riemann step $u[x, 0] = 1$ for $x \leq 0.5$, 0 otherwise

BC (Dirichlet): $u(0, t) = u_L$, $u(L, t) = u_R$ for all $t > 0$

“Since the first generation of jet airliners, there has been about a 40% improvement in aerodynamic efficiency and a 40% improvement in engine efficiency ... about half of that has come from CFD.”
(Robb Gregg, BCA Chief Aerodynamicist)

CFD Vision 2030 CFD Study: NASA Technical Reports



Eigenflow

Quantum-Enhanced CFD Solution
via Hydrodynamic Schrödinger
Equation



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Our Solution

Fig 1: 4-step mapping

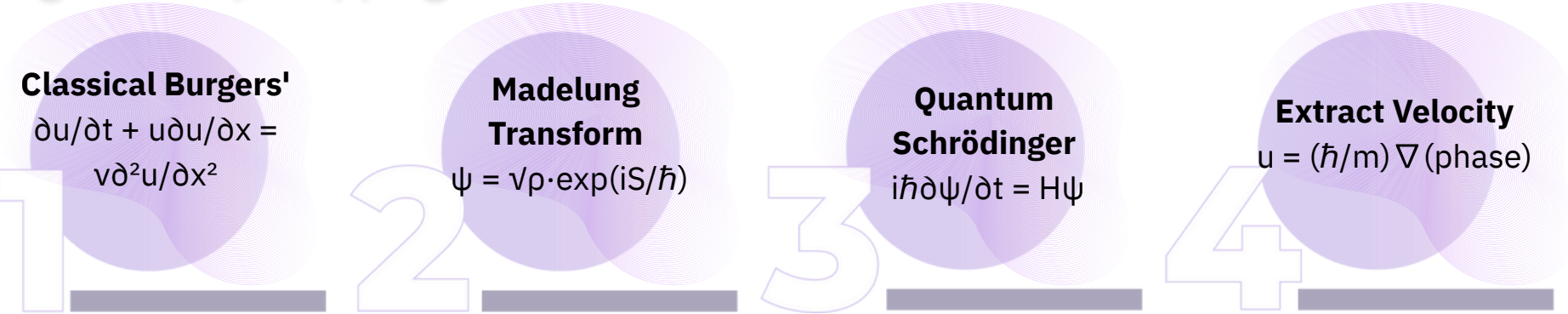


Fig 2: Quantum Circuit Representation

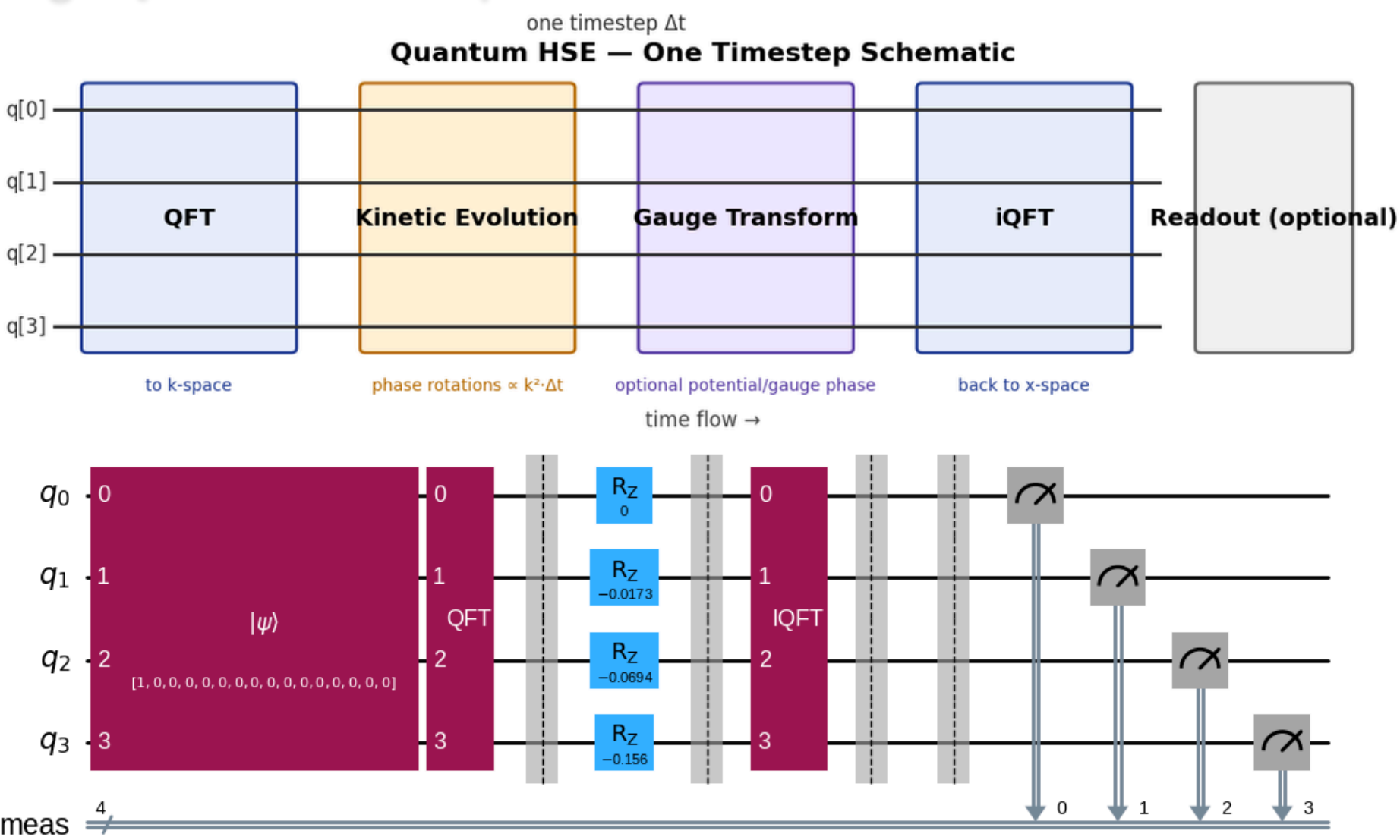


Fig 3: Tensor Network / MPS Compression

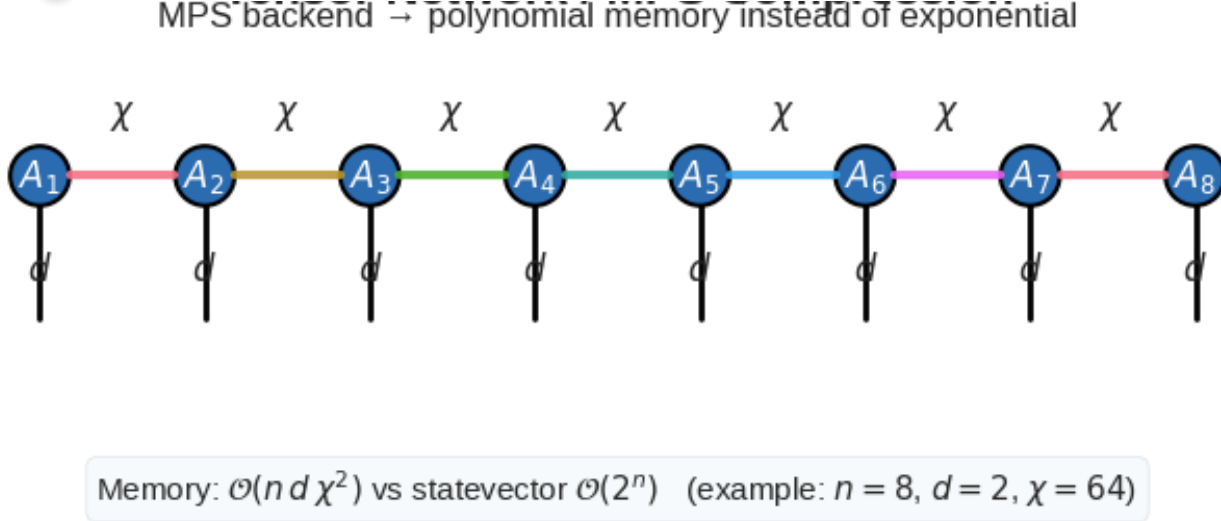
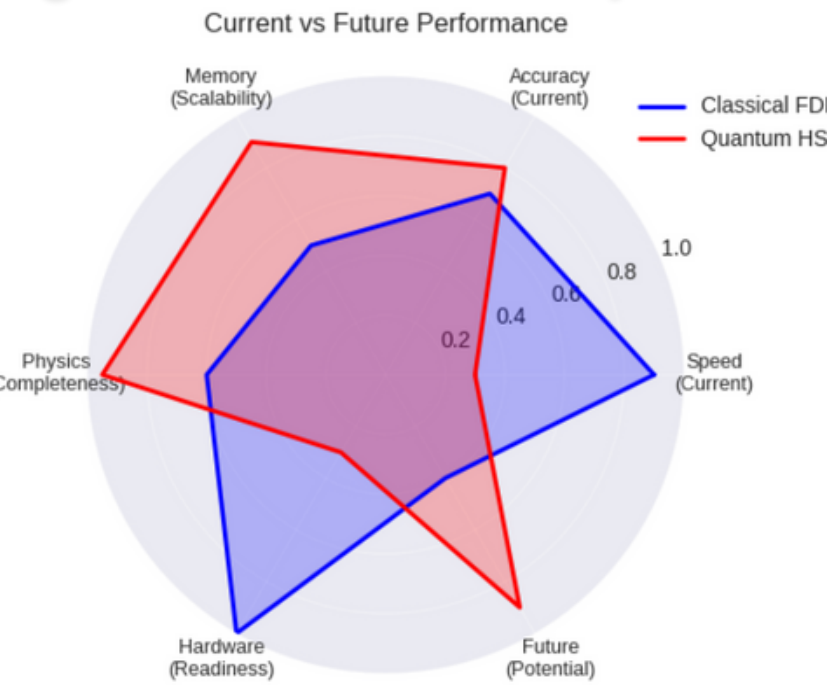


Fig 4: Five-Step Quantum Algorithm

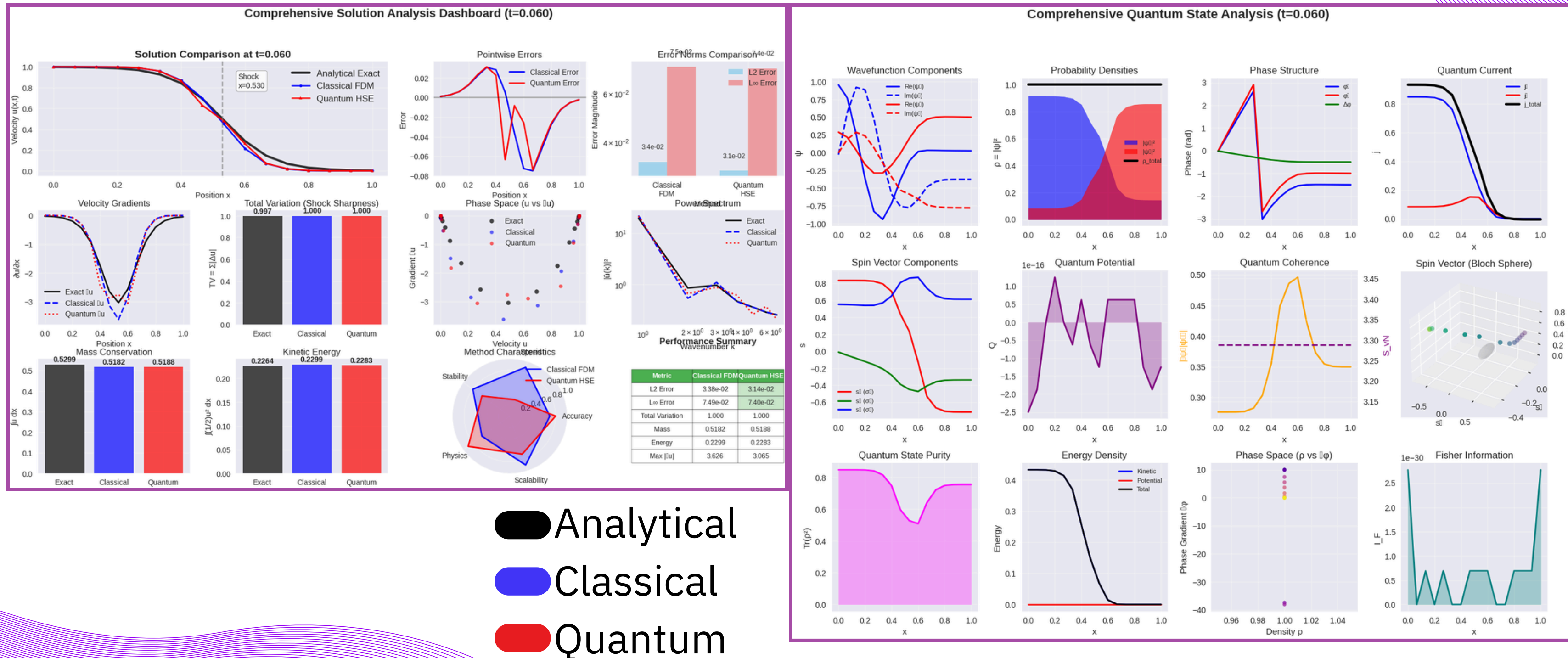


Fig 5: Hardware Feasibility



Results & impact: Solution & Quantum State Analysis Dashboard

Fig 6: Example of dashboard on quantum and classical simulations



Results

Computational Fluid Dynamics Limitations



Fig 7: Error Metrics & Accuracy

Fig 8: Burgers' Equation Solution (Simulation)

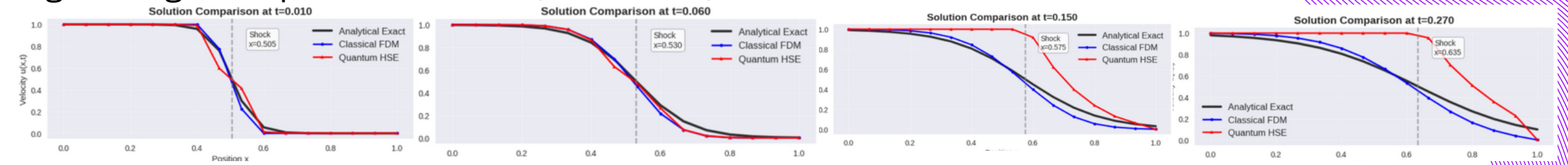


Fig 9: Burgers' Equation Solution (IBM Hardware)

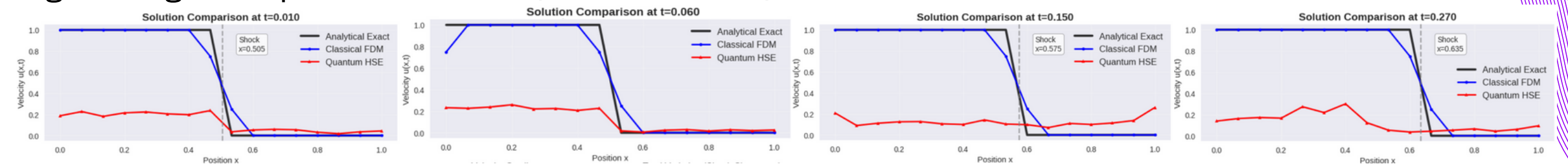


Fig 10: Velocity Gradients (Simulation)

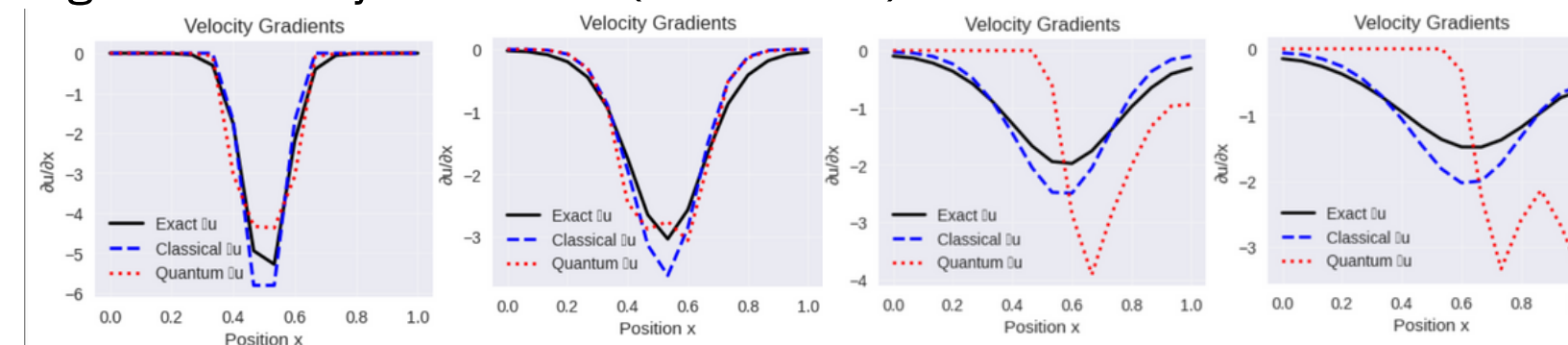
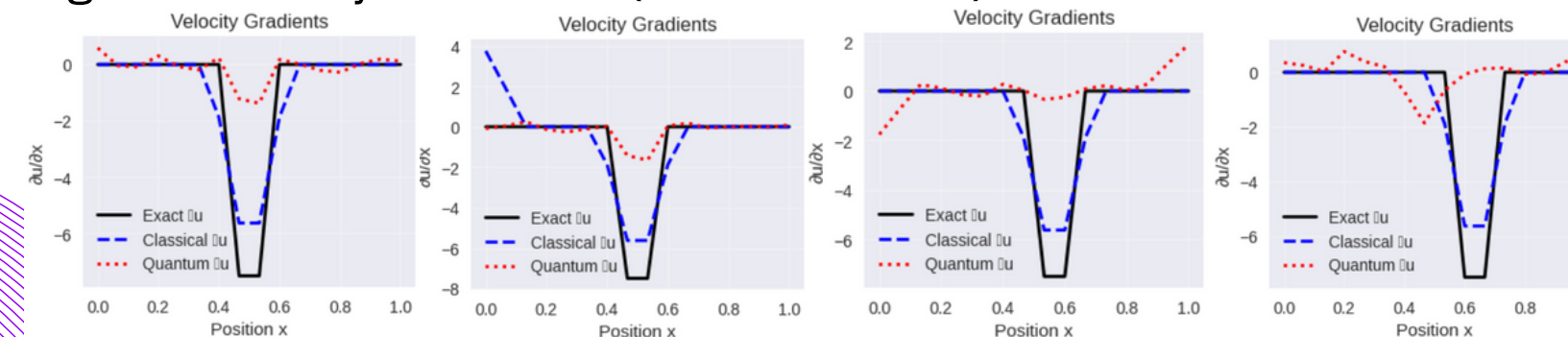


Fig 11: Velocity Gradients (IBM Hardware)



Analytical
 Classical
 Quantum

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Impact

- HSE framework with real Burgers' shocks
- Classical → quantum MPS simulation → hardware (ibm_torino), with 8-qubit feasibility and 4-qubit hardware demo
- End-to-end benchmarking, resource analysis, and hardware execution

Time (t)	Job ID	Backend	Total Meas.	Unique States	Fidelity (proxy)	Entropy (bits)	Max u	Min u	Total Variation	L2 Error	L ∞ Error	Meas. Eff.	Effective Dimensionality
0.01	d2oct7s94j0s73a2msdg	ibm_torino	1024	16	0.834	3.627	0.236	0.018	0.48	0.5603	0.8184	100.00%	10.9
0.06	d2oct7s94j0s73a2msdg	ibm_torino	1024	16	0.685	3.782	0.326	0.035	0.824	0.5866	0.8867	100.00%	12
0.15	d2oct7s94j0s73a2msdg	ibm_torino	1024	16	0.496	3.917	0.262	0.07	0.508	0.6643	0.9102	100.00%	14.1
0.27	d2oct7s94j0s73a2msdg	ibm_torino	1024	16	0.76	3.7	0.301	0.035	0.65	0.6649	0.9648	100.00%	11.2

Table 1: IBM Hardware Results

Sim: 16×60 grid/time; FDM 0.0024s vs HSE 0.0102s;

HW: 1024 shots per snapshot; entropies ~3.6–3.9 bits;

L2/L ∞ columns: Circuit depth vs decoherence tradeoff, Phase gradient extraction sensitivity, Tensor network compression limits

Future Scope: Quaternionic Liouville Hydrodynamic Schrödinger Equation (QLHSE)

Quaternionic Liouville HSE Flow

- 1. Initial Condition
 - Input Riemann step profile $u(x,0)$
- 2. Spinor Mapping
 - Represent state with two spinor components (ψ_1, ψ_2)
- 3. Spin Vector Construction
 - Compute $s=(s_1, s_2, s_3)$ from spinor amplitudes
- 4. Nonlinear Quantum Potential
 - Derive V_{nl} from gradients of sss
 - Captures quantum corrections near shocks
- 5. Gauge / Projection Step
 - Solve Poisson equation for incompressibility
 - Apply gauge transformation
- 6. Velocity Field Update
 - Extract velocity from wavefunction phase
 - Output $u(x,t)$

t	FDM L2 Error	QLHSE L2 Error	FDM Time (s)	QLHSE Time (s)
0.01	0.07955	0.517595 + 1.794269j	0.000039	7.130568
0.03	0.121885	0.506839 + 1.801007j	0.000118	21.391705
0.06	0.151251	0.502318 + 1.783788j	0.000235	42.78341
0.09	0.170623	0.575236 + 1.804060j	0.000353	64.175115
0.12	0.174866	0.571100 + 1.806782j	0.000471	85.56682
0.15	0.180979	0.574045 + 1.807153j	0.000589	106.958525
0.18	0.18835	0.574617 + 1.798833j	0.000706	128.35023
0.21	0.205178	0.635955 + 1.806651j	0.000824	149.741935
0.24	0.204353	0.647630 + 1.808124j	0.000942	171.13364
0.27	0.205234	0.646655 + 1.812003j	0.00106	192.525345

Table 2: QLHSE Simulation Results

Thank You

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6. Malik, M.R. et al. (2015) CFD Vision 2030 CFD Study, 56th HPC User Forum. https://hpcuserforum.com/presentations/virginia-april2015/NASACFD2030_HPCUser%20Forum_MalikV3.pdf.

Animation courtesy of Sora

Animation courtesy of Sora



Version	Key Technical Innovation	Implementation Details	Purpose & Physics	Resource Requirements	Validation Approach
Classical FDM Baseline <i>(Grey)</i>	Upwind finite difference scheme	Standard advection-diffusion discretization	Benchmark reference for quantum approaches	Minimal (CPU)	L ₂ error vs analytical solutions
Basic HSE <i>(Blue)</i>	Single-component Madelung transformation	$\psi = \sqrt{\rho} \cdot \exp(iS/\hbar)$, XX/YY Pauli terms	Proof-of-concept quantum CFD, diffusion-only	4 qubits, 8K shots	Direct FDM comparison
Enhanced Laplacian HSE <i>(Blue)</i>	Adds diagonal Z terms	H = XX + YY – ZZ tight-binding	Complete discrete Laplacian operator	4 qubits, 16K shots	Improved diffusion accuracy
Two-Component Spinor HSE <i>(Green)</i>	Dual 4-qubit components	Kronecker product: $\psi_{\text{full}} = \psi_1 \otimes \psi_2$	Richer fluid dynamics, component coupling	8 qubits, 16K shots	Marginal probability analysis
Vorticity-Coupled HSE <i>(Green)</i>	Cross-component XY terms	H += XY interactions	Models rotational motion, vorticity	8 qubits, 16K shots	Vorticity conservation checks
Advection-Enhanced HSE <i>(Green)</i>	ZZ Pauli terms added	H += ZZ for nonlinear advection	Captures convective transport	8 qubits, 16K shots	Shock propagation accuracy
Gaussian Initial State HSE <i>(Green)</i>	Smooth Gaussian init.	Replaces discontinuous Riemann condition	Better state preparation	8 qubits, 16K shots	Reduced initialization artifacts
QFT-Spectral HSE <i>(Green)</i>	QFT integration	Momentum-space derivative computation	Spectral accuracy for derivatives	8 qubits, 32K shots	Spectral convergence analysis
Prediction-Correction HSE <i>(Green)</i>	Multi-step quantum algorithm	QFT prediction → measurement → correction	Advanced numerical stability	8 qubits, 32K shots	Multi-step error accumulation
Cole-Hopf Validated HSE <i>(Purple)</i>	Integral transform	Exact analytical smooth solutions	Rigorous mathematical validation	8 qubits, variable shots	Cole-Hopf transform comparison
Extended Time Evolution HSE <i>(Purple)</i>	Long-term integration (nt=60)	Tests stability over long periods	Temporal stability assessment	8 qubits, 10K shots	Long-time error growth
Quantum Potential HSE <i>(Purple)</i>	Full nonlinear quantum potential	$Q = -(\hbar^2/2m) \nabla^2 \sqrt{\rho} / \sqrt{\rho}$	Complete quantum hydrodynamics	8 qubits, 100K shots	Quantum-classical transition
Incompressible Quantum Flow <i>(Purple)</i>	Divergence-free constraint	Poisson solver for projection	$\nabla \cdot \mathbf{u} = 0$ enforcement	Variable qubits	Incompressibility verification
Compressible Quantum Flow <i>(Purple)</i>	Density variation allowed	Acoustic wave propagation	Captures compressible effects	Variable qubits	Acoustic wave validation
Gauge-Transformed HSE <i>(Purple)</i>	Gauge theory integration	Phase transformations	Advanced quantum field theory	8 qubits, 10K shots	Gauge invariance checks
Spin Vector HSE <i>(Purple)</i>	3-component spin computation	$\mathbf{s} = [s_1, s_2, s_3]$ from spinor	Quantum magnetohydrodynamics	8 qubits, 10K shots	Spin dynamics validation
Poisson-Integrated HSE <i>(Purple)</i>	FFT-based Poisson solver	Spectral Poisson solution	Pressure-velocity coupling	8 qubits, 10K shots	Pressure field accuracy
MEP-Deformed HSE <i>(Purple)</i>	Higher-order deformation	γ, β parameters	Beyond-standard quantum mechanics	8 qubits, 10K shots	Deformed quantum validation
ISF Steady-State HSE <i>(Purple)</i>	Schrödinger Flow init.	Steady-state quantum prep	Optimal initial state	8 qubits, 10K shots	Steady-state verification
Multi-Condition HSE <i>(Purple)</i>	Multiple test cases	Riemann + Taylor-Green flows	Algorithmic robustness	8 qubits, 10K shots	Cross-validation analysis
Conservation-Verified HSE <i>(Purple)</i>	Conservation law tracking	Mass, energy, momentum	Physical consistency	8 qubits, variable shots	Conservation error quantification
Noise-Characterized HSE <i>(Purple)</i>	KL divergence tracking	Systematic noise model	NISQ-era robustness	8 qubits, variable shots	Noise resilience quantification



Classical

Baseline



Core HSE



Advanced HSE



Experimental /

Frontier HSE

Table x: Taxonomy of methods